

**This assignment will not be graded and is only for practice.**

1) Consider the following functions

$$f_1(x, y) = 5\sqrt{x^2 + y^2}$$

$$f_2(x, y, z) = x^2z^2 + yz + x^2y^2z^2$$

$$f_3(x, y) = \begin{cases} y \sin \frac{1}{y} + xy & \text{if } y \neq 0 \\ 0 & \text{if } y = 0. \end{cases}$$

If tangent plane exists at the origin for  $f_i$ , then enter the value of  $i$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

2) Which of the following equations represents the tangent plane to the function  $f(x, y) = 3x^2 + xy - 2y^2$  at the point (1, -1)

**1 point**

☐  $5x + 5y = 0.$

☐  $5x + 5y - z = 0.$

☐  $x + y - z = 0.$

☐  $5x + 5y + z = 0.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$5x + 5y - z = 0.$

3) Consider a function  $f(x, y) = \frac{1}{2}(xy - \frac{x}{y})$ . Let  $T$  be the tangent plane at point (1, 2). Which of the following points lie on the tangent plane  $T$ ? **1 point**

☐  $(0, 0, -\frac{5}{4})$

☐  $(1, 0, \frac{1}{4})$

☐  $(2, -4, -\frac{9}{4})$

☐  $(1, 2, -2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, -\frac{5}{4})$

$(2, -4, -\frac{9}{4})$

4) Consider a function  $f(x, y) = xy$ . Let  $T_1$  be the tangent plane at point  $(1, 1)$ ,  $T_2$  be the tangent plane at point  $(1, -1)$ . Then which of the following option is true? **1 point**

- ☐  $T_1, T_2$  intersect in only one point.
- ☐  $T_1, T_2$  intersect in infinitely many points.
- ☐  $T_1, T_2$  do not intersect.
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T_1, T_2$  intersect in infinitely many points.

5) Let  $f$  be a function from  $\mathbb{R}^2$  to  $\mathbb{R}$  and  $T$  be the tangent plane at a point. Then which of the following option(s) is (are) true. **1 point**

- ☐  $T$  is an affine subspace of the vector space  $\mathbb{R}^3$  with respect to usual addition and scalar multiplication.
- ☐  $T$  is never an affine subspace of the vector space  $\mathbb{R}^3$  with respect to usual addition and scalar multiplication.
- ☐

$T$  can be a subspace of the vector space  $\mathbb{R}^3$  with respect to usual addition and scalar multiplication.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is an affine subspace of the vector space  $\mathbb{R}^3$  with respect to usual addition and scalar multiplication.

$T$  can be a subspace of the vector space  $\mathbb{R}^3$  with respect to usual addition and scalar multiplication.

6) Consider a function  $f(x, y, z) = e^{x+y} \sin 2z$ . Which of the followings is the tangent (hyper) plane at the point  $(1, 0, \frac{\pi}{4})$ ? **1 point**

☐  $u = ex + ey$

☐  $u = ex + ey + z$

☐  $u = ex + ey + ez$

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$u = ex + ey$

7) Let  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  be a function and  $L_f$  be the linear approximation of  $f$  at point  $(a, b)$ . **1 point**  
Which of the following options is true?

☐  $L_f$  is a linear function.

☐  $L_f$  is a quadratic function.

☐  $L_f$  is a logarithmic function.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f$  is a linear function.

8) Consider a function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as  $f(x, y) = xy \tan(x) \sin^{-1}(y + \frac{\sqrt{3}}{2}) + 1$ . **1 point**  
Which of the following is the linear approximation of  $f$  at the point  $(\frac{\pi}{4}, 0)$

☐  $L_f(x, y) = x + \frac{\pi^2}{12}y$

☐  $L_f(x, y) = 1 + \frac{\pi^2}{12}y$

☐  $L_f(x, y) = 1 + y$

☐  $L_f(x, y) = \frac{\pi^2}{12}y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(x, y) = 1 + \frac{\pi^2}{12}y$

9) Consider the three functions  $f_1 = x^2 + y^2$ ,  $f_2 = xy$  and  $f_3 = x + y$ . Let  $T_1$ ,  $T_2$  and  $T_3$  be the tangent planes at points  $(-\frac{1}{2}, \frac{1}{2})$  of  $f_1$ ,  $(-1, 1)$  of  $f_2$  and  $(0, 0)$  of  $f_3$  respectively. If  $(a, b, c)$  are the intersection point of the  $T_1$ ,  $T_2$  and  $T_3$ , then find the value of  $a + 2b + c$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

10) Which of the following is the linear approximation of the function  $f(x, y) = e^x \sin y + xe^x$  at the point  $(0, 0)$  **1 point**

☐  $L_f(0, 0) = 2x + y$

☐  $L_f(0, 0) = x$

☐  $L_f(0, 0) = 2x + 2y$

☐  $L_f(0, 0) = x + y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(0, 0) = x + y$

11) Production of number of breads depends upon the quantity of raw material which **1 point** includes egg, wheat and sugar .  $x$  denotes the number of eggs,  $y$  denotes the quantity of wheat ( in kg) and  $z$  denotes the quantity of sugar (in kg) and defined as  $f(x, y, z) = xyz$ . Quality of the bread is measured by the value of linear approximation ( $L_f(x, y, z)$ ) with respect to 1 egg, 1 kg wheat, and 1 kg sugar. If value of the linear approximation is greater than 10, then quality of product is not good and if less than 10 then quality of the product is good.

Use this information to answer question.

Which of the following option(s) is (are) true?

- ☐ If 2 eggs, 5 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is good.
- ☐ If 5 eggs, 10 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.
- ☐ If 7 eggs, 8 units wheat, 9 units sugar is used to produce breads, then the quality of produced breads is good.
- ☐ If 5 eggs, 3 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If 2 eggs, 5 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is good.

If 5 eggs, 10 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

12) Consider a function  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  defined as:  $f(x, y, z) = x^2 + y^2 + z^2$ . If  $L_f(x, y, z)$  is the linear approximation of  $f(x, y, z)$  at point  $(1, 1, 0)$ , then find the value of  $\nabla L_f(x, y, z) \cdot (3, 2, 1)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

**1 point**

13) Consider a function  $f(x, y, z) = ax + by + cz$ , where  $a, b, c \in \mathbb{R}$ . If  $L_f(x, y, z)$  is the linear approximation of  $f(x, y, z)$  at the origin, then which of the following options is true? **1 point**

- ☐  $L_f(x, y, z) = 2f(x, y, z)$
- ☐  $L_f(x, y, z) = -f(x, y, z)$
- ☐  $L_f(x, y, z) = f(x, y, z)$
- ☐  $L_f(x, y, z) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(x, y, z) = f(x, y, z)$

Your score is: 0/13